COMPASS Coastal Observations, Mechanisms, and Predictions Across Systems and Scales	**COMPASS Field Sampling Protocol Chesapeake Bay Region**	
Date Created: 21 Sept 2022 Date Updated: 2 Nov 2023	Creator Name(s): A. Stearns Editor Name(s): A. Stearns	Version: 3

Vegetation Plot Installation and Sampling

Objective:

To establish replicate vegetation plots that will allow COMPASS to monitor a representative sample of plants and herbaceous ground cover over time.

Contents:

I. Experimental Design:

Each Chesapeake Bay Synoptic Site will have five (5) vegetation plots installed in each zone (Wetland, Upland and Transition). These plots will be monitored annually, during peak growth. In the Chesapeake Bay region, peak growth occurs between the July and August months.

Vegetation plots will be 1m x 1m in the Wetland Zones at GWI and MSM and be sectioned with a 10cm PVC grid system. Vegetation plots will be 2m x 2m in the Transition and Upland Zones at all sites, and in the Wetland Zone at SWH due to the nature of its vegetation.

Additional monitoring will occur in all macrophyte collars.

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Vegetation plots will be annually surveyed using the Braun Blanquet method to determine species density, abundance, and frequency. Percent cover for bare ground, coarse woody debris (CWD), basal tree stem cover and infrastructure will also be accounted for using the corresponding percent cover categories of the Braun Blanquet scoring system. Saplings over 3ft in height will not be treated as ground cover. This is consistent with TEMPEST Protocols.

In the GWI and MSM Wetland Zones, annual vegetation harvests will occur during peak growth. Vegetation from a randomly chosen 10cm grid square in each vegetation plot will be collected, dried, and weighed for biomass. Any harvested grid square will not be harvested again for five (5) years.

No vegetation harvests will be done in the Transition or Upland zones, or the SWH Wetland zone, due to the scarcity of ground cover vegetation and/or nature of the vegetation present.

Vegetation harvests from each plot will be sorted by species for live plants, and bulked for dead vegetation. Live stems will be counted, dried and weighed. Dead stems will be dried and weighed. No samples will be backlogged.

Allometry measurements will be taken when *Phragmites australis*, *Iva frutescens,* or similar vegetation is present where allometric measurements can be applied. If *Phragmites* is present, all stems within a macrophyte collar will be measured and a 50x50cm subsection in the SW corner of vegetation plots will be measured. Other shrub like vegetation will be measured within the whole plot.

II. Personal Protective Equipment: Close-toed shoes and long pants are required at all times when working at COMPASS sites. Rubber boots are strongly recommended in the Wetland and Transition zones. Chest waders are available upon request. Eye protection is available and *is required* in the marsh zones.

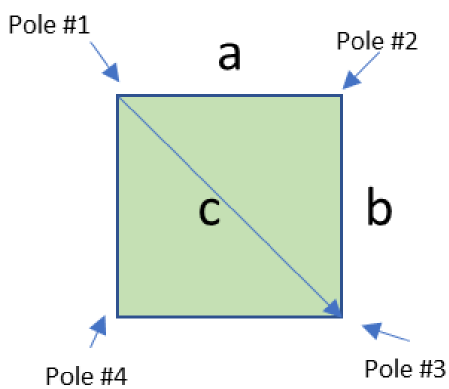
III. Installation

A. Materials:

- (3) metric tapes (at least 3M in length)
- Driveway stakes or plastic stemmed flags - 4/plot, 20/zone, or 60/site
- String
- Scissors
- PVC grids
- Vegetation Plot field notebook
- Pencil
- Sharpie

B. Procedure:

1. Designate location of the vegetation plot.
   a. Transition and Upland: Ideally, the plots should be located within the larger tree plot of the Transition and Upland zones. Using the N - S transect as your X-axis, randomly generate (x,y) coordinates and use those as the SW corner pole of the plot.
   b. Wetland: Orient the plots in front of the main catwalk >5m away to distance the plots from the sampling areas.
2. Set corner pole #1
3. Pull tape in the direction you want to the distance you want the width. *Installing plots to the cardinal direction is always preferred, if possible.*
4. Set corner pole #2
5. From corner pole #2, pull tape to the distance desired for the length, approximating a right angle, in the proposed direction of pole #3
6. Use the Pythagorean Theorem to find the hypotenuse
   Pull tape from pole #1 toward the proposed position of pole #3
7. Place pole #3 where the length of the side intercepts the length of the hypotenuse
8. Pull tapes from pole #'s 1 and 3 to find the placement of the last corner, pole #4
9. Install the PVC grid



$$a^2 + b^2 = c^2$$
$$c = \sqrt{(a^2 + b^2)}$$

IV. Sampling

A. Materials

- Vegetation Plot field notebook OR field data collection sheet
- Pencil
- Sharpie
- Varying sizes of paper bags (pre-tared bags are recommended)
- Clippers or cutting blade
- Braun-Blanquet score card
- (2) measuring tapes at least 2m long

B. Procedure

1. **Survey (all plots)**:
    - Using the Braun-Blanquet score card, conduct a vegetation survey. Document species scores in the field notebook or data collection sheet.

Include percent cover for bare ground, coarse woody debris (CWD), basal tree stem cover and infrastructure.

2. **Harvest (when applicable)**:
   - Using a random number application, select the 10cm grid square to harvest. If this happens to be a grid that has been harvested in the last five (5) years, select another grid square.
   - Label paper bags, appropriately sized for the vegetation present.
     i. Date
     ii. Site
     iii. Zone
     iv. Plot
     v. Grid #
     vi. Species or "dead"
     vii. # of stems (for live species only)
   - Clip the vegetation as close to the ground as possible.
   - Sort live stems by species. Dead stems can be combined. *This step can be done either in the field, or at the lab (if within a reasonable amount of time)*
   - Count stems by species
   - Place vegetation in the correctly labeled bag
   - Put samples in the grieve oven at 60°C for at least 72 hours
3. **Allometry (when applicable):**
   - *Phragmites australis*
     i. Macrophyte Collars
        1. Count and record all live and dead stems
        2. Measure height of all live stems
     ii. Vegetation Plots
        1. Subsection the SW corner into a 50x50cm subplot
        2. Count and record all live and dead stems
        3. Measure height of all live stems
   - *Iva frutescens*
     i. For each individual "cluster":
        1. Count the number of live stems

2. Measure diameter of stems at 40cm, up to at least five (5) stems. If there are more than five (5) stems, randomly choose the five (5) stems measured.

V. Processing

A. Materials:

- Samples
- Balance
- Weights - 1g and 50g standards
- Weigh boats
- Data collection sheet
- Pencil
- Lab gloves

B. Procedure:

1. All samples should be dried in the grieve oven at 60°C for at least 72 hours
2. Pull samples out of the oven and let them cool for about 20 minutes
3. Turn on the balance 10 minutes prior to starting
4. Make sure the balance is level
5. Weigh and record the mass of 1g and 50g standards. Be sure to weigh the standards each new day when using the balance
6. Weigh the sample. If the bag has been pre-tared, you can weigh the sample in the bag, documenting the mass of the bag on the data sheet. If not, remove the entire sample from the bag, place samples on a tared weigh boat, and record the samples' mass.
7. Record the mass on the data collection sheet or into the spreadsheet

VI. CB Site Installation Logistics

- **Goodwin Islands**
    - Installed September 2022
        - Wetland: 1m^2 vegetation plots South of the Wetland logger and MET station. Installed along a 30m transect East to west approximately 3m

apart from each other. Braun-Blanquet scores obtained. Harvested the NE 10cm grid square for processing.
  - Transition: Braun-Blanquet score was obtained from the macrophyte collars. No harvest was done. No additional plots installed for 2022.
- Installed 2023
  - Upland installed 20 July 23. Used the 20m western edge of the Tree Plot as our x-axis and used a random number app to obtain x-y coordinates to indicate the SW corner pole of each plot. Five (5) 2m^2 plots were installed
    - V1 (2,16)
    - V2 (3,2)
    - V3 (6,7)
    - V4 (15,12)
    - V5 (18,19)
  - Transition installed 21 September 23. Used the 20m western edge of the Tree Plot as our x-axis and used a random number app to obtain x-y coordinates to indicate the SW corner pole of each plot. Five (5) 2m^2 plots were installed
    - V1
    - V2
    - V3
    - V4
    - V5
  - Transition: allometry was obtained for PHAU and IVFR in both the veg plots and the macrophyte collars
  - Braun-Blanquest scores obtained for all veg and macrophyte collar plots
  - Wetland harvest was collected from the SW 10cm grid square for processing

- **Moneystump Swamp**
  - Installed September 2022
  - Wetland: 1m^2 vegetation plots West of the Wetland logger and MET station. Installed along a 30m transect East to West approximately 3m apart from each

other. Braun-Blanquet scores obtained. Harvested the NE corner 10cm grid square for processing.

- Transition: 2m^2 vegetation plots were installed. Plot locations were selected using random numbers to obtain an x,y pace. Braun-Blanquet scores observed. Harvested the SW corner 10cm grid square for processing.
- Upland: 2m^2 vegetation plots were installed. Plot locations were selected using random numbers to obtain an x,y pace. Braun-Blanquet scores observed. Harvested the ??SW corner 10cm grid square for processing.

- **GCReW**

  - Wetland: Vegetation data will be compiled from GCReW's annual plant Census
  - Transition: October 2022, 2m^2 vegetation plots were installed. Plot locations were selected using random numbers to select (x,y) coordinates from an installed N-S baseline. (Note: the tree plot had not been established at time of installation.) Braun-Blanquet scores observed. Harvested the SW 10cm grid square for processing.
  - Upland installed 18 Aug 23. Five 2m x 2m plots were installed within five of the pre-existing sixteen 5m x 5m TEMPEST Vegetation plots. Randomly selected plots are:
    - B2 - V1
    - C7 - V2
    - E7 - V3
    - G7 - V4
    - G5 - V5

    Plots were installed in the SW corner of the larger 5m x 5m TEMPEST Veg plots

- **Sweethall Marsh**
  - Installed 19 September 2023
    - Wetland: Five (5) 2m^2 plots were installed West of the sampling catwalk. Because of the nature of this wetland, the plots were installed opportunistically directly off of the catwalk. Plot 1 placed furthest to the South and then moving North numerically. BB Scores observed in Veg

plots. It was too late in the season to obtain good observations in the macrophyte collars.
    - Swamp (Upland 2023): Five (5) 2m^2 plots were installed. Used the 20m western edge of the Tree Plot as our x-axis and used a random number app to obtain x-y coordinates to indicate the SW corner pole of each plot.
    - Upland (Upland Control 2023): Five (5) 2m^2 plots were installed. Used the 15m western edge of the Tree Plot as our x-axis and used a random number app to obtain x-y coordinates to indicate the SW corner pole of each plot.
- Installed 20 September 2023
    - Transition: